

REQUEST FOR INFORMATION

Arnold Engineering Development Complex (AEDC) / Acquisition Branch (PMD), 704th Test Group (704 TG), and 704 TG/OL-AC

Extreme Loads Tire Force Machine (EL-TFM)

THIS IS A REQUEST FOR INFORMATION (RFI) ONLY.

This Request for Information (RFI) is submitted to System for Award Management (SAM) vendors under the condition that it not be released to third parties and it is understood that a funded requirement MAY NOT EXIST at this time. The requirements owner may be assessing market capabilities or defining requirements. Possible requirements derived from this solicitation for technical information may or may not result in an actual requirement, and may not be solicited under SAM, therefore approval of any future solicitation is subject to further scope determination and does not imply a guarantee of work or contract award under SAM.

This RFI is issued solely for information and planning purposes — it does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. This RFI does not commit the Government to contract for any supply or service whatsoever. Further, the Air Force is not at this time seeking proposals and will not accept unsolicited proposals. Responders are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI; all costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFP, if any is issued. If a solicitation is released, it will be synopsized on the SAM site.

1.0 AGENCY NEED

The Aerospace Survivability and Safety Office (OL-AC), located at Wright-Patterson Air Force Base (AFB), Ohio, is assigned to the 704th Test Group (704 TG) headquartered at Holloman AFB (HAFB), New Mexico. The 704 TG's mission is to be the premier provider of unique test and evaluation (T&E) capabilities relevant to tomorrow's warfighter through world-class people improving technical excellence, cost-effectiveness, and agility. The 704 TG/OL-AC operates the Landing Gear Test Facility (LGTF) at WPAFB, and the mission of the 704 TG/OL-AC and LGTF is to test various aspects of existing and prototype landing gear systems and components for the entire Department of Defense (DoD). The LGTF contains more than a dozen DoD- and world-unique test capabilities used for Research, Development, Test, and Evaluation (RDT&E) of landing gear systems and components, and maintains and operates dynamometers that are primarily used for the RDT&E of aircraft tire, wheel, and brake systems, as well as load presses, drop towers, and a Tire Force Machine (TFM).

To accomplish its current mission, the LGTF is seeking potential sources from industry interested in providing the personnel, tools, materials, supervision, logistics, training, technical support and end-end services necessary to develop the design, build, fabricate, and install an Extreme Loads Tire Force Machine (EL-TFM) system/capability at the LGTF at WPAFB capable of loading up to 250,000 lbs. Specifically, the EL-TFM project addresses a capability gap between dynamometer force testing and

static load presses. It loads the tire in a wide range of positions (vertical, yaw, and roll [or camber]), and moves the loaded surface, which is approximately 30-feet-long and approximately 5-feet-wide, at a specified rate up to approximately 6 inches/second while measuring vertical, lateral, and drag forces. Other areas investigated are the footprint forces and deformations for studies of tire wear, localized forces transmitted into the wheel, tire profile information, and quasi-static stiffness during various loading scenarios. This test system facilitates aircraft development and/or technology maturation efforts.

The contractor shall deliver an EL-TFM system that meets the requirements in the Statement of Work (SOW) – “Extreme Loads Tire Force Machine Specification (Tire/Wheel),” dated 17 September 2025. Distribution of the SOW is restricted by Distribution Statement D and interested parties must contact the Contracting Officer to confirm their eligibility to receive the SOW.

2.0 RESPONSES

2.1 Your company must be registered in SAM. To register, go to <https://www.sam.gov>. Responses are **due no later than 4 May 2026, 16:00 EST**. Responses shall be limited to 10 pages for Section 2 and submitted via e-mail only to tyler.payne.9@us.af.mil. Proprietary information, if any, should be minimized and **MUST BE CLEARLY MARKED**. To aid the Government, please segregate proprietary information. Please be advised that all submissions become Government property and will not be returned.

2.1.2 Information and Instructions:

Respondents must include the following information within their statement of capability packages. Capability packages may contain single- or double-spaced text but shall not exceed 15 single-sided pages in length.

a. Company Information

- Company Name
- Address
- Point(s) of Contact (Email and Phone Number)
- Federal CAGE Code
- Data Universal Numbering System (DUNS)
- Business size (per NAICS)
- Web Page – URL

2.1.3 Provide a Statement of Capabilities demonstrating the skills, experience, knowledge, and capabilities necessary to develop, integrate and support the EL-TFM outlined above and in the SOW. Please limit your relevant experience to the past five years to include a description of work performed, period of performance, agency/organization supported and individual points of contact (Contracting Office or Program Manager). Identify key personnel.

2.1.4 Business type (large business, small business, small, disadvantaged business, 8(a)-certified small, disadvantaged business, HUBZone small business, woman-owned small business, very small business, veteran-owned small business, service-disabled veteran-owned small business) based

upon the applicable North American Industry Classification System (NAICS) 334519, Other Measuring and Controlling Device Manufacturing. "Small business concern" means a concern, including its affiliates that is independently owned and operated, not dominant in the field of operation in which it is bidding on Government contracts, and qualified as a small business under the criteria and size standards in 13 CFR part 121. A small business concern for the purposes of this procurement is generally defined as a business, including its affiliates, averaging no more than 16.5 million dollars in annual receipts. Annual receipts of a concern that has been in business for 3 or more complete fiscal years means the annual average gross revenue of the concern taken for the last 3 fiscal years. Annual receipts of a concern that has been in business for less than 3 complete fiscal years means its total receipts for the period it has been in business, divided by the number of weeks including fractions of a week that it has been in business, and multiplied by 52. Responders are cautioned, however, that this is a general description only. Additional standards and conditions apply. Please refer to Federal Acquisition Regulation FAR 19 for additional detailed information on Small Business Size Standards. The FAR is available at <https://www.acquisition.gov>.

2.1.5 Section 2 shall contain the following:

- a. A Statement of Capabilities demonstrating the skills, experience, knowledge, and capabilities necessary to perform the work specified above and meet the SOW. Please limit your relevant experience to the past five years to include a description of the work performed, period of performance, agency/organization supported, individual point of contract (Contracting Officer/Purchasing Agent or Program Manager). Include the Contract Number(s) or Customer/Company Name.
- b. Documentation regarding experience, qualifications, and any experience working as Prime contractor.
- c. Rough Order of Magnitude (ROM) estimate for cost by major component (e.g., force measurement capabilities, table speed/size, drag load capability, mission control/data acquisition, infrastructure, etc.).
 - Kickoff and Systems Requirement Review
 - Preliminary Design Review
 - Critical Design Review
 - Facility preparation, including
 - Foundation
 - Power
 - Procurement, build, installation, integration, and testing of subcomponents, including:
 - Drive system
 - Load table system
 - Test head
 - Final Acceptance Testing
 - Training and training manuals
 - Program management, systems engineering, and master scheduling and documentation

- d. Describe your approach for this project to include which major components, if any, you would subcontract and how you would manage the integration.
- e. Describe your company's experience with developing test machines capable of measuring normal force (F_z), lateral force (F_y), drag/friction force (F_x), and torsional moments (M_x , M_y , and M_z) on aircraft tires and applying camber and yaw angles to aircraft tires (or related projects).
- f. Describe your company's experience developing test machines that can accommodate aircraft-sized tires and the associated table speed and length of these test machines (or related projects).
- g. Describe your company's experience developing test machines capable of measuring and applying drag loads to aircraft-sized tires and brakes.
- h. What is the minimum and maximum dollar value for a project which your firm would routinely consider submitting for a proposal?
- i. What information, not currently provided, does your company expect to need to provide the EL-TFM components? What assumptions would have to be made to bid on this effort?
- j. What is the estimated milestone schedule for the design, procurement, installation, and test for the EL-TFM system?

3.0 INDUSTRY QUESTIONS

Contracting Officers may or may not choose to meet with potential offerors. Such discussions would only be intended to get further clarification of potential capability to meet the requirements, especially any development and certification risks.

4.0 QUESTIONS

Questions regarding this announcement shall be submitted in writing by e-mail to the Contracting Officer. Verbal questions will NOT be accepted. The Government does not guarantee that questions will be answered. The Government does not guarantee that questions received after **20 April 2026, 16:00 EST** will be answered.

5.0 SUMMARY

THIS IS A REQUEST FOR INFORMATION (RFI) ONLY to identify sources that can provide an EL-TFM capability that meets the requirements above and the SOW. The information provided in the RFI is subject to change and is not binding on the Government. The Air Force has not made a commitment to procure any of the items discussed, and release of this RFI should not be construed as such a commitment or as authorization to incur cost for which reimbursement would be required or sought. All submissions become Government property and will not be returned.

6.0 REFERENCES

1. SOW – Extreme Loads Tire Force Machine Specification (Tire/Wheel), dated 17 September 2025